How Should a Personal Agent Store Its Facts? Granularity, Compute, and Robustness in Parametric Memory

Mahdi Ghodsi **Megha Sri Satya Sai Devineni** **Chandra Shekhar Pandey**
Advanced Micro Devices, Inc.

Abstract

A personal language-model agent must keep a user's facts somewhere: as prompt context at query time or encoded in adapter weights. When memory is partitioned, both approaches require selection: one retrieves text into context and the other activates a memory partition. The open design question is then not whether to select but how finely to partition what is selected. We study this on a 4B instruction-tuned model over a fixed 200-fact user memory, holding per-adapter rank constant at 16 and varying the number of partitions across three adapter-construction regimes. Recall from adapters compiled by a hypernetwork falls steeply as partitions coarsen, from 0.602 at 20 facts per adapter to 0.116 at 200, against a 0.050 base rate. Gradient-trained adapters given exactly 249 optimizer updates each decline similarly (0.746 to 0.354, $p = 8.4 \times 10^{-17}$). When instead every partition receives equal per-fact exposure at 83 epochs, original-question recall remains 0.740–0.773 without a monotonic decline, while recall on paraphrased questions still deteriorates with coarser partitions (0.564 to 0.387); the direct question-form interaction is 0.160 (author-clustered 95% CI [0.105, 0.212]). Granularity therefore strongly affects parametric memory, but the relationship depends on the construction regime and training exposure rather than on fixed adapter capacity alone, and robustness and recall respond differently. We measure two in-context points on the same items: among the five matched operating points, placing all 200 facts in every prompt reaches 0.862 original and 0.834 paraphrased recall at 131.4$\times$ the prompt tokens of the parametric arms per query, with top-5 retrieval intermediate on both accuracy (0.729) and cost (4.67$\times$). No mechanism dominates every measured quality and cost axis; we report what each one costs.

1 Introduction

An agent that runs on personal hardware has to remember its user: that your daughter's recital is Thursday, that your build server is the one in the closet, and several hundred other things no pretrained model was ever told. The engineering question is where that knowledge lives.

The usual framing is a dichotomy: put the facts in the context window and retrieve them, or bake them into the weights and dispense with retrieval. That dichotomy misses the selection cost of partitioned parametric memory. A system with one adapter per topic still selects: instead of retrieving text into the prompt, it retrieves a *memory partition* and activates it, so only the delivery changes, tokens or weights. Once either representation is partitioned, both approaches introduce a selection problem, and the interesting variable becomes granularity: how much of a user's memory should one adapter hold? Coarse partitioning is cheap (one adapter, no routing, minimal storage); fine partitioning can improve accuracy when construction work per adapter is constrained, but costs bank storage, construction, and routing.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: SLMs for Agentic Systems, Paris, France, 2026.

We fix a user memory of 200 facts, fix per-adapter rank at 16, and vary how many partitions the memory is split across: 20, 40, 100, and 200 facts per adapter, a bank of 10 adapters down to a single monolithic one. Holding per-adapter rank fixed might appear to make the sweep a pure capacity measurement. Our results show that interpretation is too strong. How the adapters are built matters as much as how the facts are divided: under a fixed per-adapter budget of optimizer updates, recall on original questions falls steeply as partitions coarsen, but when every partition instead receives equal per-fact gradient exposure, that decline is not observed. What does survive equalizing exposure is the paraphrase penalty (Figure 1). We then price these organizations against what they cost, and against two in-context alternatives on identical items: top-5 retrieval, and placing the entire 200-fact memory in every prompt.

Our contributions:

- **A controlled granularity $\times$ construction-budget study.** A four-point 20/40/100/200 facts-per-adapter sweep over one fixed 200-fact memory at fixed per-adapter rank, in three construction regimes (compiled, gradient at a fixed budget of 249 optimizer updates, and gradient at fixed per-fact exposure), with paired item-level tests in each (§4.1).
- **A separation of original-form recall from robustness.** Under equal per-fact exposure, the resolved coarse-granularity penalty on original-question recall is no longer observed, while the paraphrase penalty remains in every regime we measured (§4.1).
- **The systems accounting.** Five operating points on the same items, full context, top-5 retrieval, compiled and gradient adapters, and the memoryless base, priced against construction time, bank storage (28.3–283.2 MB for the same facts), routing, and prompt tokens per query (§4.2, §4.3).

2 Related work

Several prior works motivate partitioned adapter memory; we claim no priority for the architecture itself. We hold per-adapter rank constant and vary the number of partitions, but construction regime and exposure change how that sweep reads, so it is not on its own a capacity measurement.

Capacity of a single adapter. Back et al. [2026] sweep data volume at fixed rank and report a rank-dependent saturation point beyond which "performance degrades sharply"; Hurtado [2026] internalizes 1–99 documents and confirms "a capacity trend (rank must grow with corpus size)." Both vary rank along their ladder or hold the adapter count at one; our axis is the number of partitions with per-adapter rank held constant. Tan et al. [2026] find 1.7–2.75 bits per parameter and that placement matters more than count, which motivates our MLP-side target module; Xu et al. [2026] fit a power law in rank and sequence length, measuring loss not free-form recall, and Allen-Zhu and Li [2024] establish the pretraining-scale analogue.

Partitioned and routed adapter memory. Back et al. [2026] also include a partition arm, under the constraint that is the crucial difference from ours: configurations are compared "under an identical parameter budget," one rank-32 module against eight rank-4, so per-adapter capacity *shrinks* as they partition. We hold it fixed and let total storage grow, the constraint a deployment faces when adapters are a fixed served artifact. Diao et al. [2026] already build per-atom micro-adapters with a query router, so we claim no architectural priority; their rank is fixed at 8, the atom count K is a training curriculum with inference fixed at $K = 8$, and they name the granularity axis as future work. Zheng et al. [2026] distill each context into its own adapter and report that "accumulating contexts weakens an adapter's ability to preserve context-specific knowledge," but their cumulative baseline updates sequentially, confounding forgetting with holding more content; ours are independent.

Compiling context into weights. Charakorn et al. [2026] train a hypernetwork converting a document into a LoRA in one forward pass, our compiled-memory generator; their rank ablation holds content fixed while varying capacity, the inverse of our sweep. Eyuboglu et al. [2025] sweep cartridge size at a fixed corpus and find LoRA scales poorly to large cartridges relative to prefix tuning; corpus size is never varied. Pletenev et al. [2025] vary the number of injected facts but measure damage to *pre-existing* knowledge rather than recall of the injected facts.

Personal memory. Tan et al. [2024] attach a per-user parametric module and find parametric and non-parametric memory complementary rather than substitutable. Deng [2026] and Li [2026] argue that style and factual recall respond oppositely to weight edits; we take that substrate split as given and ask how to organize the factual half. We adopt the small-language-model definition of Belcak et al. [2025]: a model below roughly 10B parameters fitting on a consumer device.

3 Experimental design

Setting and data. A single user with 200 personal facts, served by one shared 4B base model small enough to run on hardware the user controls. The facts are TOFU [Maini et al., 2024], config `forget10_perturbed`, fictitious author biographies designed to avoid overlap with real-world factual knowledge. We do not assume zero pretraining exposure, since TOFU predates the base model; the base arm is the empirical check, and its 0.050 recall indicates little of the evaluated information is recoverable without the memory mechanism. Each item is a question, a gold answer, and a human-written paraphrase of both. The corpus is blocked by author in groups of 20 rows, verified free of cross-block name leakage; we treat the 200 facts of ten authors as one user's memory.

The granularity ladder. We partition the same 200 facts into 20, 40, 100, and 200 facts per adapter, corresponding to 1, 2, 5, and 10 author blocks per adapter and banks of 10, 5, 2, and 1 adapters. Partitions are deterministic and author-aligned, blocks packed in ascending author order, so no author is split across adapters. The compiled generator never chunks a document, so every adapter is emitted at rank 16 with identical tensor count and size, verified across all 18. Per-adapter capacity is therefore fixed, but total system capacity is not: finer partitioning uses more adapters and more total bytes, the cost we account for in §4.2. A silent truncation at coarse granularity would look exactly like "big adapters forget," so the build compares the tensor the encoder received against the tokenizer output for every document and fails if they differ; all four granularities pass.

Controlling the gradient construction budget. Optimizer updates and per-fact exposure cannot both be held fixed while partition size varies, so we measure both. In the **fixed-update** regime every adapter takes exactly 249 updates, which is 83 epochs at 20 facts per adapter but under ten at 200. In the **equal-exposure** regime every partition trains for 83 epochs, so updates grow with partition size (249, 415, 1079, 2075). Both use rank-16 LoRA on `down_proj`, learning rate 2×10^{-4} on a linear schedule with no warmup, fused AdamW, batch size 1 with gradient accumulation 8, bf16, over identical deterministic partitions. The fixed-update arm resets all RNGs to seed 42 immediately before LoRA initialization and we verified an identical initial draw across all 18 adapters, so initialization does not vary with build order. Termination is verified per adapter by reading `global_step` back, which equalled 249 in every case. The equal-exposure conditions are trained independently, each with its own linearly annealed scheduler horizon, so that sweep is not one learning trajectory and its points are not checkpoints along a common run. These are two resource protocols, not an isolation of exposure: fixed updates equalize work per adapter while changing record exposure and total bank work; equal epochs hold record passes fixed while changing updates, scheduler horizon, bank size, and total stored parameters.

Gradient training representation. Each gradient record is one chat-formatted original TOFU question as the user turn and its original answer as the assistant turn. Thus all original evaluation questions and answers occur verbatim in gradient training, while TOFU's paraphrased questions and answers are excluded. We use ordinary causal-language-model loss over the full rendered conversation, without assistant-only masking. Compiled adapters instead receive the corresponding answer statements as documents. Original-question recall therefore measures recall of a seen question-answer form; the paraphrase condition measures transfer to new wording.

Arms. **Base** is the unmodified model with no memory. **Compiled** adapters are generated by a document-to-LoRA hypernetwork [Charakorn et al., 2026] in a single forward pass. **Gradient** adapters are trained with standard LoRA fine-tuning [Hu et al., 2021] on the same partitions. **Top-5 retrieval** places the five highest-scoring retrieved facts in the prompt [Lewis et al., 2020]. **Full context** places all 200 facts in every prompt in a fixed order, with no ranking, filtering, or top-k step; it is therefore the no-selection reference point rather than a retrieval arm. All arms answer the same

items with the same prompt template, grader, threshold, and generation settings; the full-context arm alone required a server configured with a 16,384-token context window to hold its prompts.

Metric. Free-form generation, graded by discriminative token recall against the gold answer with a perturbed distractor as a control; an item is correct at a threshold of 0.7. We report free-form correctness rather than a forced choice because in preliminary work the forced-choice metric concealed failures free-form generation exposed: a model can rank the gold answer above a distractor while unable to produce it. For item i, let K_i be the content tokens shared by its original and paraphrased gold answers, excluding tokens also shared by every factually wrong TOFU perturbation. A generation is correct when it contains at least 70% of K_i. Original and paraphrased questions use the same K_i. For 19 items this filtering leaves K_i empty; we exclude them from every paired comparison rather than count a grader abstention as model error, leaving $n = 181$ of 200. The wrong-answer floor is 0.0166.

Routing. Oracle routing sends each question to the partition holding its fact, isolating the memory question from routing. Predicted routing uses a unigram content-word overlap router over the partition text, with no access to labels or gold answers, and is the same router for both adapter families. TOFU names the author in the question, making routing unusually easy (0.990 top-1 at 20 facts per adapter). Headline numbers are oracle-routed.

Statistics. The same 200 items recur at every point on the ladder, so pooling cells as independent observations would shrink intervals for reasons unrelated to evidence. Comparisons along the curve are therefore paired at the item level and tested with exact McNemar on discordant pairs. Granularity-curve aggregate intervals use an item-clustered bootstrap; single-arm binomial intervals use Wilson intervals.

Serving. vLLM 0.24.0 on ROCm 7.2.3, one MI300X GPU, Qwen3-4B-Instruct-2507 base [Qwen Team, 2025a,b]; gradient adapters use rank 16, $\alpha = 32$, and learning rate 2×10^{-4} throughout. The operating-point comparison of §4.3 uses a gradient arm matched to the compiled arm on optimizer steps rather than epochs, 249 steps per adapter (83 epochs at 20 facts); it is a separate experiment from the granularity regimes above and its numbers are not interchangeable with them. Every evaluated parametric adapter passed a log-probability activity gate (§4.4).

Artifacts. Code, frozen per-item results, generated tables and figures, and reproduction instructions are available at `https://github.com/Mahdi-CV/personal-agent-memory-granularity`.

4 Results

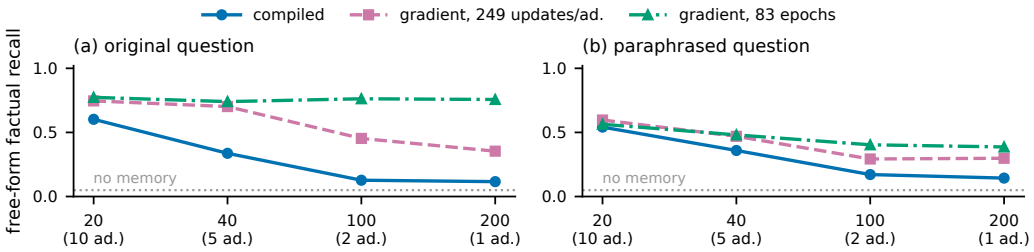
4.1 Granularity across three construction regimes

Figure 1 gives the main result. The three regimes are different resource settings, not three estimates of one curve, and we report them separately throughout.

Compiled adapters. Recall falls from 0.602 at 20 facts per adapter to 0.116 at 200, against a base rate of 0.050: the steepest granularity dependence of the three regimes, front-loaded, with 100 to 200 unresolved ($p = 0.804$). That floor is a property of the compiled generator at this rank and budget, not of adapters in general: the gradient regimes reach higher recall at the same partition sizes and rank.

Gradient adapters at a fixed update budget. Giving every adapter exactly 249 optimizer updates reproduces the decline, 0.746 to 0.354, with 77 items lost and 6 gained end to end ($p = 8.4 \times 10^{-17}$). This budget is per adapter, not across the bank, since finer partitions contain more adapters; under it, finer partitions substantially improve factual recall.

Gradient adapters at equal per-fact exposure. Holding exposure rather than updates fixed, original-question recall remains 0.740–0.773 across the tested granularities, none of the three paired comparisons against 20 facts per adapter reaching significance (all $p \geq 0.42$): the monotonic



		compiled		gradient, 249 updates		gradient, 83 epochs			bank	
facts/ad.	ad.	orig.	para.	orig.	para.	orig.	para.	upd.	MB	comp. build s
20	10	0.602	0.541	0.746	0.597	0.773	0.564	249	283.2	7.06
40	5	0.337	0.359	0.702	0.470	0.740	0.481	415	141.6	3.80
100	2	0.127	0.171	0.453	0.293	0.762	0.403	1079	56.6	1.74
200	1	0.116	0.144	0.354	0.298	0.757	0.387	2075	28.3	1.14
$20 \to 200$, p		1.4e−23	5.2e−19	8.4e−17	9.0e−11	0.77	2.1e−5			

Figure 1: **Granularity depends on the construction regime.** The x axis is facts per adapter over one fixed 200-fact memory, per-adapter rank held at 16 and 28.3 MB, built three ways; oracle-routed, $n = 181$ paired items. (a) On original questions the compiled and fixed-update regimes decline as partitions coarsen while the equal-exposure regime does not; (b) on paraphrased questions all three degrade. Bank build seconds are for the compiled regime; the table's last row is the end-to-end $20 \to 200$ paired test in each regime.

original-question decline is not observed here. That is an unresolved comparison, not evidence of equivalence, which would require an equivalence test we did not perform; and coarser adapters reach it only by taking up to 2075 updates against 249.

Robustness behaves differently. The paraphrase penalty does not dissolve under equal exposure: recall on paraphrased questions still falls from 0.564 to 0.387 (44 lost, 12 gained, $p = 2.1 \times 10^{-5}$), and the same comparison is resolved in the fixed-update and compiled regimes (Figure 1). Whatever extra exposure supplies to a coarse partition, it restores original phrasing more readily than robustness to rephrasing.

Direct interaction and author clustering. The equal-exposure dissociation is also resolved directly: the 20-to-200 coarsening penalty is 0.160 larger for paraphrased than original questions. An author-clustered bootstrap over the 10 authors gives a 95% interval of [0.105, 0.212] (exact author-level sign-flip $p = 0.0039$). Author-level endpoint tests preserve the other central readings: compiled and fixed-update original recall decline ($p = 0.002$ and $p = 0.002$), equal-exposure original recall does not ($p = 0.578$), and its paraphrase recall declines ($p = 0.008$). We retain item-level tests for this evaluated memory and use the author analysis as a clustered sensitivity check.

4.2 What finer partitioning costs

At fixed 28.3 MB per adapter, finer partitioning raises total bank storage from 28.3 MB to 283.2 MB (10× for the identical 200 facts) and compiled construction from 1.14 s to 7.06 s (Figure 1). Two serving-time costs stay flat: resident memory, since one partition is active per query, and prompt length, since the selected memory arrives as weights.

Gradient construction follows the resource constraint: fixed-update bank time falls from 2027.6 s at 20 facts per adapter to 243.4 s at 200 as the bank shrinks from ten adapters to one. Under 83 equal-exposure epochs, bank time stays near 1562.4–1687.9 s across granularities because the bank processes roughly the same total number of records.

The third cost is routing, and it can be wrong. Our lexical router reaches 0.990 top-1 accuracy at 20 facts per adapter, misrouting 2 of 200 items, but we claim no headline routing tax: oracle-versus-predicted differences are at most 0.022 absolute, below the engine's reproducibility floor. Multi-

Table 1: Operating points for the same 200-fact memory. Prompt tokens recur per query, as a ratio to the 57.6-token parametric prompt; bank MB and bank build seconds are one-time totals across the ten-adapter g20 bank (the text gives per-adapter build seconds), and zero means no adapter-construction cost, not zero indexing or serving infrastructure.

memory organization	orig. q	para. q	prompt tok.	bank MB	bank build s
Base (no memory)	0.050	0.072	1×	0	0
Full context (all 200 facts)	0.862	0.834	131.4×	0	0
Top-5 retrieval	0.729	0.613	4.67×	0	0
Compiled, 20 f/ad.	0.602	0.541	1×	283.2	7.06
Gradient, 20 f/ad.	0.779	0.597	1×	283.2	1731.5

LoRA decoding under concurrency is not bitwise reproducible at temperature zero here; repeating the coarsest arm changed 56 generations and 1 graded outcome.

4.3 Operating points: in-context and parametric memory

Table 1 compares memory organizations on the same items at the same 4B scale; none wins everywhere.

Full context. At this corpus size all 200 facts fit directly in context, the highest recall among these five points, 0.862 original and 0.834 paraphrased (156 / 181 correct, Wilson 95% CI [0.804, 0.905]), with no resolved paraphrase penalty (12 lost, 7 gained, $p = 0.359$). It uses 7,567.6 prompt tokens per query, 131.4× the parametric prompt and 28.1× top-5 retrieval; these are prompt-token measurements, and we infer no latency or throughput ratios from them. Since this arm answers the same items with the same grader, questions and grader alone do not account for the parametric paraphrase decline, though storage is not established as its sole mechanism.

Top-5 retrieval is intermediate on both axes, 0.729 original recall at 4.67× the parametric prompt, with zero adapter cost in our accounting: not a claim that retrieval is free, since it needs an index, a scorer, and the corpus.

Gradient-trained adapters at 20-fact granularity reach the highest original-question recall among the matched parametric points (0.779), but take 173.2 s per adapter to build against 0.706 s compiled; trained on all 200 facts at once with a wider target-module set they reach 0.906 but fall to 0.547 under paraphrase. **Compiled adapters** are below both on raw recall and 245.3× cheaper to construct here.

Figure 1 and Table 1 use independently trained 20-fact gradient banks. The figure's seed-controlled bank was rebuilt jointly for the granularity ladder and scores 0.746; the table retains the earlier step-matched bank built for routing and systems accounting, scoring 0.779. Both use rank 16, `down_proj`, and 249 updates, but they are different adapter artifacts and evaluation passes.

Paraphrase robustness we report within each arm, not as a ranking. Rewording costs the gradient arm significantly (retention 0.660, $p = 3.8 \times 10^{-5}$) and the top-5 retrieval arm significantly (0.742, $p = 0.003$), while neither the compiled arm's drop (0.817, $p = 0.061$) nor the full-context arm's (0.923, $p = 0.359$) is resolvable at this sample size. On a common-correct paired test every pairwise comparison between the parametric and top-5 retrieval arms is non-significant (all $p \geq 0.31$), so we do not rank them here. As a control, on questions unrelated to the stored facts all arms sit near the floor (base 0.119, compiled 0.091, retrieval 0.068).

4.4 Reliability of parametric memory in serving

A successful adapter load is not evidence that the intended parametric memory is active: in vLLM 0.24.0 an adapter with mismatched tensor names loads with HTTP 200 while applying *zero* weights, flattening the curve toward the base rate, a silent failure indistinguishable from a negative result. Treating bitwise-identical logprobs as failure, every evaluated parametric adapter passed a log-probability activity gate against a base-versus-base noise floor.

5 Limitations

Routing is easy here. TOFU names the author in the question, so selecting the partition is close to a keyword match, whereas real personal memory refers to entities obliquely and has overlapping partitions. Our routing numbers are an upper bound and the tax unresolved, not zero.

One dataset, one scale, one adapter configuration. A single 4B base, one target-module set, rank 16, one learning rate, and for the compiled arm one compiler. We did not sweep adapter capacity, so we make no claim about how these curves move with rank, and prior work indicates the saturation point does move [Back et al., 2026]. TOFU is fictitious and biographical; we rely on the base arm rather than a non-exposure guarantee, and its paraphrases are human rewrites, not natural queries.

Facts and entities co-vary. Partitions are author-aligned, so 20, 40, 100, and 200 facts also mean 1, 2, 5, and 10 authors per adapter. The coarse-partition penalty may therefore include entity-binding or cross-entity interference, not only the number of stored records.

The construction budget is part of the result. Optimizer budget and exposure materially change the relationship between granularity and original-question recall, so a granularity sweep does not measure adapter capacity on its own and we do not report one as if it did. We tested two points in that space, not the space itself; intermediate budgets, schedules, and optimizers are unmeasured, and the equal-exposure conditions also differ in scheduler horizon, which we did not separate from exposure. We control initialization within the fixed-update ladder but do not estimate variance across independent training seeds. The independently trained 20-fact banks differ by 0.033, underscoring that run-level variance is unmeasured.

Our comparison is not a general verdict on in-context memory. We charge top-5 retrieval for prompt tokens, not its index, corpus, or retriever infrastructure, and prompt-token ratios do not establish latency or throughput under load. At 200 facts the whole memory fits in a 16k context and performs best; we did not vary corpus scale, so we do not identify where full context becomes inferior or infeasible, and Hurtado [2026] reports the opposite ordering on a 15-document corpus. Our full-context prompts used a fixed order, confounding position with item difficulty. Finally, we measure factual recall, not what an adapter carries *well*: TOFU holds voice constant, a clean fact and unusable style benchmark.

Compiler scope. All compiled documents were consumed without truncation, but the coarse documents are longer than the compiler's meta-training chunks. The compiled curve can therefore reflect both adapter granularity and compiler distribution shift; the gradient regimes provide a second construction mechanism but do not identify which explains the compiled arm's steeper decline.

6 Conclusion

Personal memory can be supplied as prompt tokens or encoded in adapter weights, with different selection requirements and operating costs. Among the five matched operating points, placing the whole memory in context reaches 0.862 original and 0.834 paraphrased recall at 131.4$\times$ the parametric prompt tokens per query under this prompting policy. Among parametric organizations, fine partitioning helps when construction updates are constrained and aids robustness to rewording in every regime, but under equal per-fact exposure the monotonic original-question decline is not observed. Which organization fits depends on the constrained resource, not on adapter capacity alone.

References

Zeyuan Allen-Zhu and Yuanzhi Li. Physics of language models: Part 3.3, knowledge capacity scaling laws. *arXiv preprint arXiv:2404.05405*, 2024.

Seungju Back, Dongwoo Lee, Naun Kang, Taehee Lee, S. K. Hong, Youngjune Gwon, and Sungjin Ahn. Understanding LoRA as knowledge memory: An empirical analysis. *arXiv preprint arXiv:2603.01097*, 2026.

Peter Belcak, Greg Heinrich, Shizhe Diao, Yonggan Fu, Xin Dong, Saurav Muralidharan, Yingyan Celine Lin, and Pavlo Molchanov. Small language models are the future of agentic AI. *arXiv preprint arXiv:2506.02153*, 2025.

Rujikorn Charakorn, Edoardo Cetin, Shinnosuke Uesaka, and Robert Tjarko Lange. Doc-to-LoRA: Learning to instantly internalize contexts. *arXiv preprint arXiv:2602.15902*, 2026.

Youwang Deng. Substrate asymmetry in user-side memory: A diagnostic framework. *arXiv preprint arXiv:2606.11712*, 2026.

Xingjian Diao, Wenbo Li, Yashas Malur Saidutta, Avinash Amballa, Lazar Valkov, and Srinivas Chappidi. Doc-to-atom: Learning to compile and compose memory atoms. *arXiv preprint arXiv:2606.12400*, 2026.

Sabri Eyuboglu, Ryan Ehrlich, Simran Arora, Neel Guha, Dylan Zinsley, Emily Liu, Will Tennien, Atri Rudra, James Zou, Azalia Mirhoseini, and Christopher Ré. Cartridges: Lightweight and general-purpose long context representations via self-study. *arXiv preprint arXiv:2506.06266*, 2025.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.

Joan Figuerola Hurtado. Data quality over capacity: Internalizing documents into LoRA adapters for closed-book QA. *arXiv preprint arXiv:2607.21861*, 2026.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. *arXiv preprint arXiv:2005.11401*, 2020.

Bojie Li. User as engram: Internalizing per-user memory as local parametric edits. *arXiv preprint arXiv:2606.19172*, 2026.

Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C. Lipton, and J. Zico Kolter. TOFU: A task of fictitious unlearning for LLMs. *arXiv preprint arXiv:2401.06121*, 2024.

Sergey Pletenev, Maria Marina, Daniil Moskovskiy, Vasily Konovalov, Pavel Braslavski, Alexander Panchenko, and Mikhail Salnikov. How much knowledge can you pack into a LoRA adapter without harming LLM? *arXiv preprint arXiv:2502.14502*, 2025.

Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025a.

Qwen Team. Qwen3-4B-Instruct-2507. Hugging Face model repository, 2025b. URL `https://huggingface.co/Qwen/Qwen3-4B-Instruct-2507`.

Kaizhen Tan, Heqing Du, and Yang Feng. How many bits can an adapter write? measuring the capacity and memorization of parameter-efficient fine-tuning. *arXiv preprint arXiv:2607.21351*, 2026.

Zhaoxuan Tan, Qingkai Zeng, Yijun Tian, Zheyuan Liu, Bing Yin, and Meng Jiang. Democratizing large language models via personalized parameter-efficient fine-tuning. *arXiv preprint arXiv:2402.04401*, 2024.

Ziwen Xu, Haiwen Hong, Linsong Yu, Benglei Cui, Longtao Huang, Hui Xue, and Ningyu Zhang. How LoRA remembers? a parametric memory law for LLM finetuning. *arXiv preprint arXiv:2605.30260*, 2026.

Ziyang Zheng, Zeju Li, Xiangyu Wen, Jianyuan Zhong, Junhua Huang, Lei Chen, Mingxuan Yuan, and Qiang Xu. Context distillation as latent memory management. *arXiv preprint arXiv:2605.28889*, 2026.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and the three contribution bullets (§1) each name the measurement that backs them: the granularity sweep in three adapter-construction regimes (§4.1), the storage/construction/routing accounting (§4.2), and the five-way operating-point comparison including full context and top-5 retrieval (§4.3). Scope is stated in the same sentences: one 4B model, one 200-fact TOFU memory, per-adapter rank fixed at 16. Where the abstract says the monotonic original-question decline is not observed under equal per-fact exposure, the text states that this is an unresolved comparison rather than an equivalence result (§4.1). The abstract closes by declining to name a winner, which matches §4.3.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §5 is four dedicated paragraphs: routing is unusually easy on TOFU and our routing numbers are an upper bound; a single base model, target-module set, rank and learning rate, with no capacity sweep, so no claim about how these curves move with rank; the construction budget is part of the result, so a granularity sweep does not measure adapter capacity on its own and we tested two points in the budget/exposure space rather than the space itself, with scheduler horizon not separated from exposure in the equal-exposure arm; and the in-context comparison charges top-5 retrieval for prompt tokens but not for index, corpus, or retriever infrastructure, does not vary corpus scale (citing a paper that reports the opposite ordering at a smaller corpus), used one fixed full-context ordering so position is confounded with item difficulty, and measures factual recall only, since TOFU holds authorial voice constant. The routing tax is explicitly reported as unresolved rather than zero, because it falls below the engine's own reproducibility floor (§4.2).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper is purely empirical. It states no theorem and derives no analytical result.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: §3 gives the dataset and exact config (TOFU `forget10_perturbed`), the partition rule (author-aligned 20-fact blocks packed in ascending order, which is deterministic), the base model, rank, learning rate, α, target module, optimizer, batch size and gradient accumulation, the grading rule and threshold, the pairing and test procedure, and the serving stack with versions. Both gradient construction regimes are specified separately: the fixed-update regime (exactly 249 optimizer updates per adapter, RNGs reset to seed 42 immediately before LoRA initialization, termination verified by reading `global_step` back) and the equal-exposure regime (83 epochs per partition, trained independently each with its own scheduler horizon, giving 249–2075 updates). Two controls needed to reproduce the result honestly are also described: the per-document truncation gate (§3) and the log-probability activity gate (§4.4). Caveats stated in the paper: the serving stack is not bitwise reproducible under concurrency (§4.2), so exact per-item replication is not expected, and the full-context arm required a server configured with a 16,384-token context window rather than the configuration used by the other arms.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code?

Answer: [Yes]

Justification: The camera-ready release includes the experiment scripts, frozen per-item results, generated tables and figures, environment details, and step-by-step reproduction instructions. The public TOFU dataset and Qwen base model are cited at their points of use.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details needed to understand the results?

   Answer: [Yes]

   Justification: §3 specifies rank 16, the MLP-side target module (`down_proj`), the learning rate (2×10^{-4}) and α (32) held constant across the ladder, the oracle and predicted routing procedures, the grading threshold of 0.7, and $n = 181$ of 200 after dropping ungradable items pairwise. The two granularity regimes are distinguished explicitly, since optimizer updates and per-fact exposure cannot both be held fixed while partition size varies: fixed-update (249 updates per adapter) and equal-exposure (83 epochs per partition). The operating-point comparison of §4.3 uses a separate step-matched gradient arm (249 steps per adapter), and the paper states that its numbers are not interchangeable with the granularity regimes.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Because the same 200 items recur at every point on the ladder, comparisons are paired at the item level and tested with exact McNemar on discordant pairs. The end-to-end $20 \rightarrow 200$ paired p-value is reported for each of the three construction regimes, on both original and paraphrased questions, in the table inside Figure 1, and lost/gained counts are given in the text (§3, §4.1). Interval types are distinguished: granularity-curve aggregate intervals use an item-clustered bootstrap that resamples items and carries all of an item's measurements together, whereas single-arm binomial intervals, including the full-context 95% interval in §4.3, are Wilson intervals. Figure 1 plots point estimates without error bars, with significance carried by the paired tests in the accompanying table rather than by interval overlap. The paper also states where a difference is *not* resolvable rather than reporting it as an effect: the 100-to-200 compiled step ($p = 0.804$), every equal-exposure original-question comparison against 20 facts per adapter (all $p \geq 0.42$, described as unresolved rather than as equivalence), and the routing tax, which sits below the engine's reproducibility floor.

8. **Experiments compute resources**

   Question: Does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

   Answer: [Yes]

   Justification: §3 states one MI300X GPU running vLLM 0.24.0 on ROCm 7.2.3 with a 4B base model. Construction cost, which is the dominant compute term and is itself a reported result, is given per adapter in the prose of §4.3 (0.706 s compiled against 173.2 s gradient-trained), whereas Table 1 reports the corresponding ten-adapter bank totals (7.06 s compiled against 1731.5 s gradient-trained); per-granularity compiled build seconds appear in the table inside Figure 1. Evaluation cost is not separately tabulated, and we report prompt tokens rather than wall-clock time for the in-context arms, since the full-context arm ran under a different server configuration and its runtime is not comparable to the others.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The study uses a public synthetic benchmark of fictitious author biographies and a public base model. No human subjects, no personal or private data, and no annotator labour are involved.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: The work is a measurement study of where a personal agent should store facts. It introduces no new model, capability, or artifact for release. We note that the setting concerns private user data, but our results are about how finely a memory is partitioned and how the adapters are constructed, not about whether stored information can be recovered or removed; the paper makes no privacy or unlearning claim, and we do not consider the measurement itself to carry a distinct societal risk.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

    Answer: [N/A]

    Justification: No data, model, or scraped corpus is released with this submission, and the adapters trained here encode only fictitious biographies from a public benchmark, so there is no misuse surface to guard.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: Every asset used is cited at the point of use in §3: TOFU [Maini et al., 2024] for the data, Qwen3-4B-Instruct-2507 base [Qwen Team, 2025a,b] for the model, the document-to-LoRA hypernetwork [Charakorn et al., 2026] for the compiled generator, LoRA [Hu et al., 2021] for the gradient arm, and retrieval-augmented generation [Lewis et al., 2020] for the retrieval arm. Note honestly: individual license names and versions are not printed in the manuscript; all assets used are publicly released and used within their published terms, and license details will accompany the code release.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: This submission releases no new asset. The adapters built for the study are experimental artifacts, not a contributed model.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation?

    Answer: [N/A]

    Justification: No crowdsourcing and no human subjects. The paraphrases used in the evaluation are the human-written paraphrases shipped with the TOFU benchmark, not collected by us.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No study participants and no human-subjects research, so no review was required.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: [Yes]

Justification: Language models are the object of study, and every such use is described in §3: a Qwen3-4B-Instruct-2507 base as the model under test, a document-to-LoRA hypernetwork as the compiled-memory generator, gradient-trained LoRA for the gradient arm, and the same base prompted with retrieved or full-corpus context for the in-context arms. Grading is a deterministic token-recall rule against the gold answer, not an LLM judge.